

MONTEVIDEO, MN, USA

WMO: 726553

Lat: 44.969N

Lon: 95.710W

Elev: 1034

StdP: 14.16

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04951

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.6	-8.9	-21.8	1.7	-11.6	-17.2	2.2	-8.3	31.5	9.9	27.3	12.2	6.9	300	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	90.3	73.4	87.9	72.7	84.1	70.3	77.4	86.4	75.3	83.8	73.3	81.2	11.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.7	135.0	83.2	72.5	125.2	80.9	70.4	116.4	79.0	41.7	85.8	39.4	83.9	37.6	82.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.1	23.0	19.8	-19.3	95.8	6.2	2.7	-23.8	97.7	-27.4	99.3	-30.9	100.8	-35.4	102.8	
			DB												
			WB	-19.7	81.1	5.9	2.8	-23.9	83.1	-27.4	84.8	-30.7	86.4	-35.0	88.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.4	14.4	18.3	31.0	45.4	58.5	68.6	72.4	69.4	62.5	48.9	33.6	20.0
	DBStd	22.86	13.82	13.18	13.01	10.39	9.07	6.62	5.57	5.35	8.47	9.61	11.56	12.20
	HDD50	4401	1104	887	598	204	27	0	0	0	7	140	504	930
	HDD65	7818	1569	1307	1053	590	243	37	8	20	146	507	943	1395
	CDD50	2719	0	0	10	67	290	559	695	602	382	104	10	0
	CDD65	662	0	0	0	3	40	146	238	157	71	7	0	0
	CDH74	6209	0	0	3	61	512	1368	2200	1278	688	98	1	0
	CDH80	1972	0	0	0	15	179	447	746	350	209	26	0	0
	WSAvg	8.9	9.4	9.6	9.9	10.8	10.1	8.5	6.9	6.6	7.7	9.1	9.3	9.1
	PrecAvg	25.7	0.9	0.9	1.5	2.5	3.3	4.0	3.4	3.1	2.7	2.0	1.3	0.7
Precipitation	PrecMax	39.1	3.1	3.3	5.9	6.5	8.9	10.5	9.2	9.1	6.1	6.7	5.4	3.3
	PrecMin	13.1	0.0	0.0	0.1	0.2	0.9	0.5	0.1	0.4	0.4	0.0	0.0	0.0
	PrecStd	6.7	0.8	0.9	1.1	1.3	1.9	2.4	1.9	1.8	1.7	1.7	1.2	0.7
	PrecStd	6.7	0.8	0.9	1.1	1.3	1.9	2.4	1.9	1.8	1.7	1.7	1.2	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.0	53.9	71.6	81.2	91.0	93.3	93.0	90.8	90.7	83.6	69.7	50.4
		MCWB	38.6	47.8	55.3	60.3	65.4	72.5	76.8	76.8	71.6	62.9	53.9	43.8
	2%	DB	39.5	45.2	61.4	73.2	84.3	88.7	90.2	87.7	85.6	75.1	61.4	43.4
		MCWB	35.1	40.0	52.2	55.4	63.7	70.9	76.4	75.2	69.4	59.6	49.4	37.3
	5%	DB	36.3	39.3	54.4	68.2	80.6	84.4	87.8	83.7	81.7	70.0	54.7	38.7
		MCWB	33.1	35.9	47.9	53.1	61.9	68.4	75.0	72.6	66.9	57.3	46.5	34.7
	10%	DB	33.7	36.4	47.9	63.2	74.6	81.7	83.9	81.3	78.6	63.9	48.5	35.9
		MCWB	31.6	33.8	43.2	50.6	59.0	66.7	72.4	70.0	65.2	53.9	42.3	33.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	47.8	58.5	62.4	71.8	76.9	81.3	80.1	75.6	66.7	56.5	45.1
		MCDB	45.7	52.3	65.4	75.7	83.7	87.8	87.1	87.6	84.0	78.0	66.1	48.1
	2%	WB	35.9	40.1	53.2	58.3	67.2	74.3	78.1	76.8	72.6	62.5	51.5	38.3
		MCDB	40.2	44.8	60.8	69.9	78.2	84.2	86.8	84.8	81.5	71.4	58.8	42.8
	5%	WB	33.5	36.7	48.0	54.8	64.1	71.7	76.1	74.2	69.9	58.8	47.3	35.4
		MCDB	35.9	39.7	55.0	65.3	74.0	81.4	84.8	81.0	78.1	66.9	54.5	38.5
	10%	WB	31.2	33.8	43.1	51.8	61.2	69.1	74.2	72.0	67.1	55.1	43.0	32.8
		MCDB	33.2	36.4	47.9	61.1	71.7	79.1	82.3	79.0	75.2	62.7	48.9	35.4
Mean Daily Temperature Range	5% DB	MDBR	16.1	16.9	16.6	19.8	20.9	19.6	19.8	19.9	22.2	19.4	16.4	14.8
		MCDBR	19.1	20.2	25.3	30.7	29.2	24.3	22.3	23.2	26.8	28.9	25.3	19.7
		MCWBR	15.8	15.9	16.8	16.4	13.7	11.3	12.1	13.1	13.2	15.6	16.1	15.4
	5% WB	MCDBR	17.5	18.9	24.3	25.9	23.5	20.6	20.6	20.5	22.4	24.1	23.8	18.3
		MCWBR	14.7	15.1	16.6	15.8	13.5	11.8	12.7	13.4	13.0	15.1	16.1	14.7
Clear-Sky Solar Irradiance	taub		0.264	0.275	0.307	0.360	0.383	0.394	0.398	0.405	0.365	0.318	0.282	0.262
	taud		2.452	2.427	2.445	2.340	2.338	2.366	2.338	2.299	2.394	2.513	2.531	2.468
	Ebn at Noon		276	292	296	285	281	278	275	268	271	272	267	265
	Edh at Noon		22	27	31	37	39	38	39	39	32	25	20	19
All-Sky Solar Radiation	RadAvg		571	908	1208	1504	1730	1927	2073	1717	1331	860	560	415
	RadStd		66	104	117	139	129	129	72	141	99	116	52	44

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page